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Assignment 2: Hadoop Multi-Node Cluster Installation

Set up a Hadoop multi-node cluster (NameNode + DataNode + ResourceManager + NodeManager) on a single Linux machine using **Docker images** from the Big Data Europe (`bde2020`) project. Each service runs as an isolated Docker container on a shared bridge network, simulating a real distributed cluster.

Steps

Step 1 — Prerequisites

Ensure Docker is installed:

```
_multi_node$ docker --version
Docker version 28.3.3, build 980b856
```

Create Docker Files

```
services:
  # ── NameNode (Master - HDFS)
  namenode:
    image: bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8
    container_name: namenode
    hostname: namenode
    restart: always
    ports:
      - "9870:9870"   # HDFS NameNode Web UI → http://localhost:9870
      # port 9000 (HDFS IPC) is internal-only between containers
    environment:
      - CLUSTER_NAME=big-data-cluster
    env_file:
```

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```
    - .env
  volumes:
    - namenode_data:/hadoop/dfs/name
  networks:
    - hadoop-net

# ── DataNode (Worker - HDFS)
datanode:
  image: bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8
  container_name: datanode
  hostname: datanode
  restart: always
  ports:
    - "9864:9864"   # DataNode Web UI → http://localhost:9864
  env_file:
    - .env
  volumes:
    - datanode_data:/hadoop/dfs/data
  networks:
    - hadoop-net
  depends_on:
    - namenode

# ── ResourceManager (Master - YARN)
resourcemanager:
  image: bde2020/hadoop-resourcemanager:2.0.0-hadoop3.2.1-java8
  container_name: resourcemanager
  hostname: resourcemanager
  restart: always
  ports:
    - "8088:8088"   # YARN Resource Manager Web UI → http://localhost:8088
  env_file:
    - .env
```

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```
    networks:
      - hadoop-net
    depends_on:
      - namenode
      - datanode

  # ── NodeManager (Worker - YARN)
  nodemanager:
    image: bde2020/hadoop-nodemanager:2.0.0-hadoop3.2.1-java8
    container_name: nodemanager
    hostname: nodemanager
    restart: always
    ports:
      - "8042:8042"   # NodeManager Web UI → http://localhost:8042
    env_file:
      - .env
    networks:
      - hadoop-net
    depends_on:
      - namenode
      - datanode
      - resourcemanager

volumes:
  namenode_data:
  datanode_data:

networks:
  hadoop-net:
    driver: bridge
```

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Step 2 — Start the Cluster

Docker will automatically pull the images from Docker Hub (first run only) and start all 4 containers.

```
⊗ /assignment2_hadoop_multi_node$ docker compose up
 [+] Running 4/0
  ✔ Container namenode         Created                                                    0.0s
  ✔ Container datanode         Created                                                    0.0s
  ✔ Container resourcemanager  Created                                                    0.0s
  ✔ Container nodemanager      Created                                                    0.0s
 Attaching to datanode, namenode, nodemanager, resourcemanager
 namenode         | Configuring core
 namenode         |  - Setting fs.defaultFS=hdfs://namenode:9000
 namenode         |  - Setting hadoop.http.staticuser.user=root
 namenode         | Configuring hdfs
 namenode         |  - Setting dfs.webhdfs.enabled=true
 namenode         |  - Setting dfs.replication=1
 namenode         |  - Setting dfs.namenode.name.dir=file:///hadoop/dfs/name
```

Step 3 — Verify the Cluster

●	assignment2_hadoop_multi_node	-	-	-
●	namenode	6400cfc76b6f	bde2020/hadoop-namenode:2.0.0-hadoop3.:	9870:9870
●	datanode	39f112f6b5cd	bde2020/hadoop-datanode:2.0.0-hadoop3.2.	9864:9864
●	resourcemanager	404ae579cfdc	bde2020/hadoop-resourcemanager:2.0.0-ha(	8088:8088
●	nodemanager	dcb236168adb	bde2020/hadoop-nodemanager:2.0.0-hadoo	8042:8042

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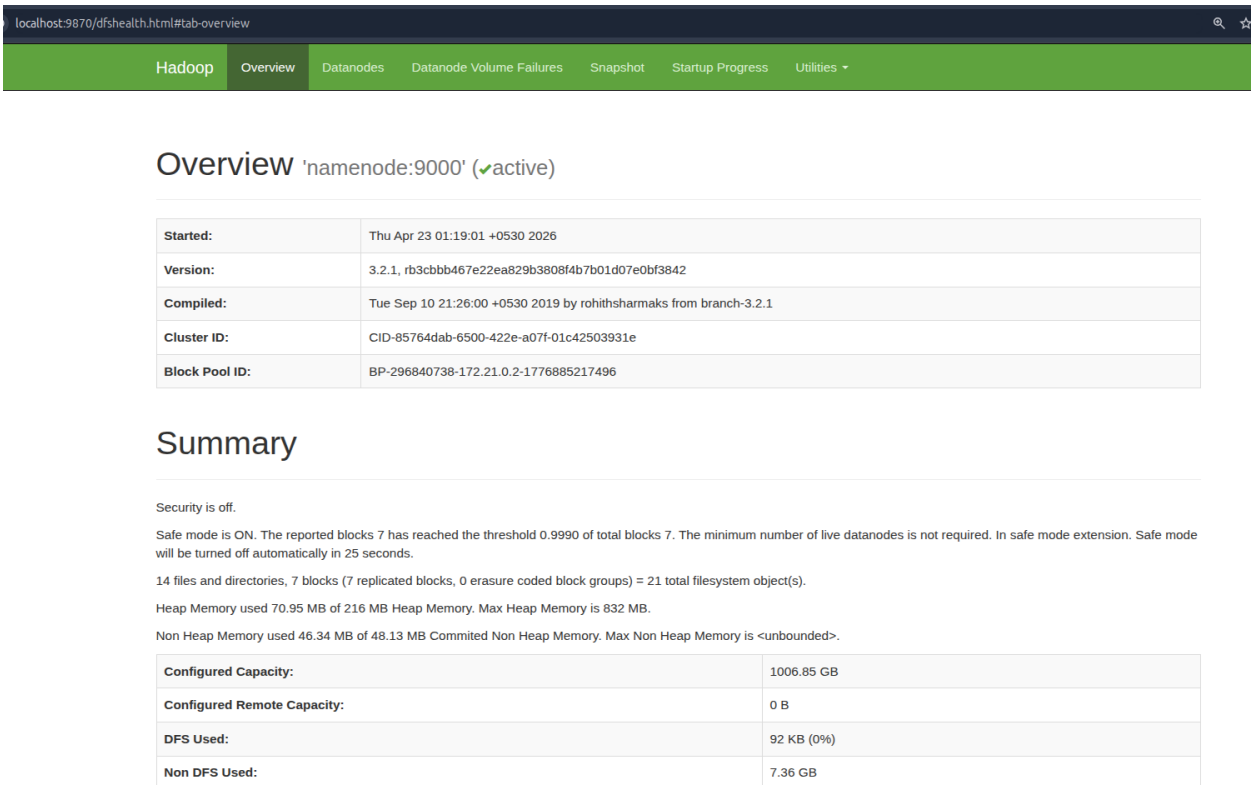
Step 5 — Web Interfaces

Service	URL
HDFS NameNode UI	http://localhost:9870
YARN Resource Manager	http://localhost:8088
DataNode UI	http://localhost:9864
NodeManager UI	http://localhost:8042

The NameNode UI at `localhost:9870` shows **1 Live Node** (the `datanode` container) confirming multi-node operation.